

K. Aditya Bharadwaj Nemana

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Education

- Sept 2021 – present **McGill University**, PhD in Mechanical Engineering – Montreal, Canada
GPA: 3.9/4.0
- **Relevant Courses:** Turbulence, Numerical Analysis I, and Matrix Computations.
 - **Thesis:** Matrix-free preconditioners for Jacobian-free Newton-Krylov Methods applied to turbulent flows.
 - **Awards:** Recipient of the McGill Engineering Doctoral Award (MEDA).
- Sept 2019 – Feb 2022 **McGill University**, MSc in Mechanical Engineering – Montreal, Canada
GPA: 4.0/4.0
- **Relevant Courses:** Computational Aerodynamics, Computational Gas Dynamics, Turbomachinery and Propulsion, Dynamics of Combustion, and Finite Element Methods in CFD.
 - **Thesis:** Perturbation Signal Tracking Techniques for Unsteady Shock Wave Flows
 - **Awards:** Recipient of the M. P. Paidoussis Prize for the best MSc thesis in the Department of Mechanical Engineering, McGill University, in 2022.
- July 2014 – May 2018 **National Institute of Technology Kurukshetra**, B.Tech in Mechanical Engineering – Kurukshetra, India
GPA: 9.45/10.0
- **Relevant Courses:** Fluid Dynamics, Thermodynamics, Numerical Mathematics, Calculus, and Power Generation.
 - **Project:** Reduced-order model for hypersonic flow simulation in an axisymmetric intake with a conical forebody, supervised by the Defense Research and Development Laboratory (DRDO), India.

Academic Experience

- Sept 2021 – April 2025 **Teaching Assistant**, McGill University – Montreal, Canada
Courses: Fluid Mechanics 2, Fundamentals of Computer Programming using Python.
- Responsible for conducting tutorials, grading assignments, and providing feedback to students.
 - Assisted with the development of course materials and resources.
- July 2018 – May 2019 **Research Associate**, Indian Institute of Technology (IIT) Kanpur – Kanpur, India
Project: Development of a Standalone, Liquid fueled Miniature Power Generation System at Advanced Combustion and Acoustics Laboratory.
- Worked on the design and development of a miniature power generation system using liquid fuel.
 - Conducted Computational Fluid Dynamics (CFD) analyses using ANSYS-FLUENT and ICEM-CFD to evaluate the performance and efficiency of the system.
 - Optimized the heat sink design, resulting in a publication in the Journal of Energy Resources Technology.

Professional Experience

- April 2025 – present **Communications Assistant A**, International Student Services (ISS), McGill University – Montreal, Canada

- Responsible for redesigning and maintaining the ISS website to improve user experience and access to information.
 - Maintaining WCAG 2.0 AA (Level AA) and SGQRI 008 2.0 conformity using Editoria11y Accessibility Checker and WAVE.
 - Assisted with the planning and execution of ISS events and initiatives.
- June 2024 – present **Accessibility Commissioner**, Post-graduate Students Society (PGSS), McGill University – Montreal, Canada
- Active member of McGill University’s Subcommittee for People with Disabilities and its website development team.
 - Organized events and workshops to raise awareness about accessibility issues faced by grad students and promote inclusivity.
 - Developed an accessibility survey for graduate students and postdocs at McGill University.
 - Responsible for ensuring that the upcoming new PGSS website conforms with WCAG 2.2 Level AA standards.
- Jan 2025 – present **SCOPE Fellow**, Office of E-IDEA, McGill University – Montreal, Canada
- Member of the Social Impact, Community Building, and Outreach Partnerships in Engineering (SCOPE) team.
 - Responsible for designing modules to help graduate students integrate the principles of Equity, Diversity, and Inclusion (EDI) into engineering design.
 - Responsible for integrating the designed modules into a self-paced course website.
 - Providing a neurodivergent perspective to the module and website design through my lived experiences.
- Sept 2023 – April 2025 **IT Assistant A**, AV Loans Center, McGill University – Montreal, Canada
- Responsible for processing loans and returns of AV equipment and resources in the Faculty of Education.
 - Provided technical support and assistance to users of AV equipment and resources.
- May 2024 – Aug 2024 **Summer Intern**, Office of E-IDEA, McGill University – Montreal, Canada
- Researched and compiled various approaches to integrating the principles of Equity, Diversity, and Inclusion (EDI) into engineering design.
 - Created, collected, and maintained a database of engineers from marginalized communities worldwide who are underrepresented in the colonial engineering curriculum.
 - Created, collected, and maintained a database highlighting ingenious engineering innovations by Indigenous peoples of Canada.

Volunteering Experience

- Mar 2022 – present **Events Coordinator**, Graduate Engineering Equity Committee (GEEC), McGill University – Montreal, Canada
- Organized monthly EDI-centric events such as movie nights, guest speaker talks, panel discussions, etc., on behalf of GEEC.
 - Managed event logistics including creating registration forms, posters, room bookings, advertising, and collecting feedback.
- Sept 2014 – May 2018 **Events Coordinator**, Shiksha, National Institute of Technology Kurukshetra, – Kurukshetra, India
- Taught children from Kindergarten to Grade 2 in Mathematics, Science, and English.
 - Organized science fairs and cultural festivals to promote overall growth among students, and coached children to participate in various competitions such as debates and quizzes.

- Organized and participated in annual fundraising campaigns for the benefit of the children.

Skills

Programming/Software: Proficient with Python, C++, Git, MATLAB, bash, Markdown, WMS, MS Office, semantic HTML, Audacity, Canva, Blender, Hugo CMS

Mathematics: Good understanding of differential equations, calculus, and linear algebra

Accessibility Tools: Editoria11y Accessibility Checker, WAVE Browser Extension, Siteimprove Accessibility Checker, NVDA, PDF Accessibility Checker - PAC

Languages: English, Hindi, Telugu, and Odiya

Publications

- 2025 **On the Disturbance Tracking Methods for Shock Wave Reflection Studies**

Nemana, K. A. B., Timofeev, E.

- 2019 [10.1063/5.0286049](#) (*AIP Conference Proceedings, Vol. 3315, No. 1, pp. 030001*)
Development of a Standalone, Liquid-Fueled Miniature Power Generation System

Bharadwaz, N. K. A., Jain, N., Arghode, V. K.

[10.1115/1.4044764](#) (*Journal of Energy Resources Technology, Vol. 142, No. 4, pp. 042004*)